



Technical Progress Report

Project name:	Biped robot		
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1. Executive Summary

This report summarizes the technical progress of the biped robot project in the past week. The team successfully completed the motor connection and confirmed the correct pin positions. However, the 3D-printed structural parts failed to fit the motors properly due to insufficient tolerance design, so assembly is suspended until revised parts are printed next week.

2. Objectives & Scope

Complete motor wiring and confirm pin definitions for the biped robot.
Assemble mechanical structures and install motors into 3D-printed parts.
Test basic motor operation and prepare for subsequent motion control.

3. Technical Work Completed

3.1 Electrical Connection & Pin Definition

- Successfully connected all motors of the biped robot.
- Confirmed and fixed the pin positions for control and power supply.
- Verified basic power conduction and circuit connectivity.

3.2 Mechanical Structure Inspection

- Inspected the size and assembly fit of 3D-printed structural parts.
- Checked the matching between motor appearance and installation slots.

3.3 Hardware Preparation

- Prepared motor drivers, main controller, and power supply modules.
- Completed pre-debugging of the electrical system.

4. Observations & Findings

- Motor wiring and pin definition are accurate and reliable.
- The 3D-printed parts lack sufficient tolerance for motor installation.
- Motors cannot be fixed into the preset positions correctly.
- Mechanical assembly cannot be carried out until parts are re-printed.

5. Challenges & Roadblocks

- 3D printing design did not consider motor size tolerance, leading to installation mismatch.
- Structural assembly is blocked, delaying the overall integration progress.
- No effective temporary adjustment for the ill-fitting printed parts.

6. Next Steps & Recommendations

- Redesign 3D-printed structural parts with proper installation tolerance and clearance.
- Re-print the modified mechanical parts as soon as possible.
- Complete motor installation and structural assembly next week.
- Conduct motor drive tests and preliminary motion debugging after assembly.

7. Conclusion

This week, the electrical system of the biped robot made smooth progress, with successful motor connection and pin confirmation. Mechanical assembly was suspended due to insufficient tolerance in 3D-printed parts. The overall progress is under control, and assembly and debugging will be finished next week after re-printing.